# Causal Differentiating Concepts: Interpreting LM Behavior via Causal Representation Learning

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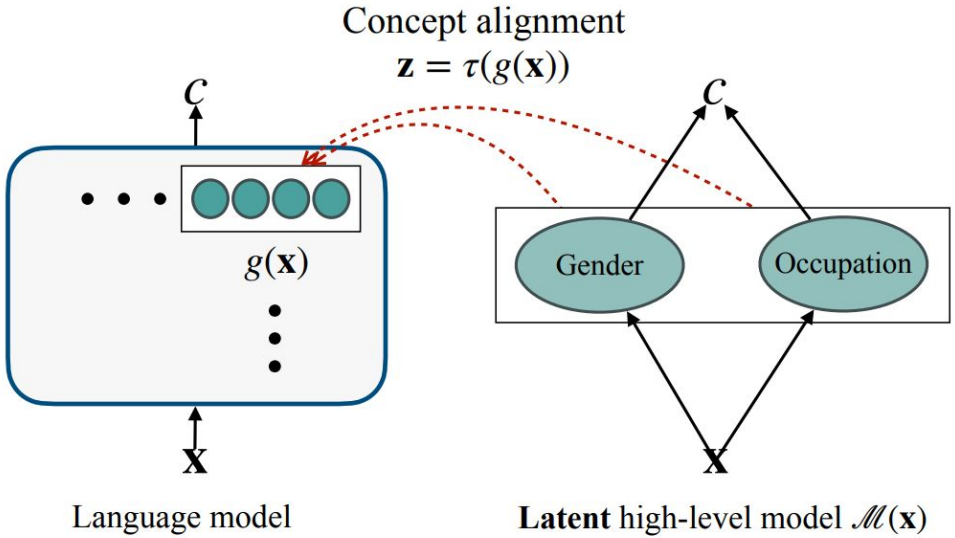
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# Causal Representation Learning

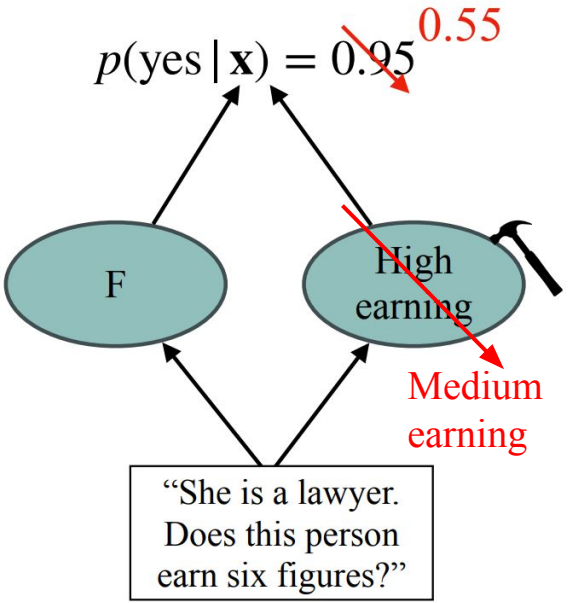
- Learning to map from high-dimensional features/embeddings ***X*** to a set of latent causal variables ***Z***

- Helps to understand how changing one variable affects the output decision ***C***



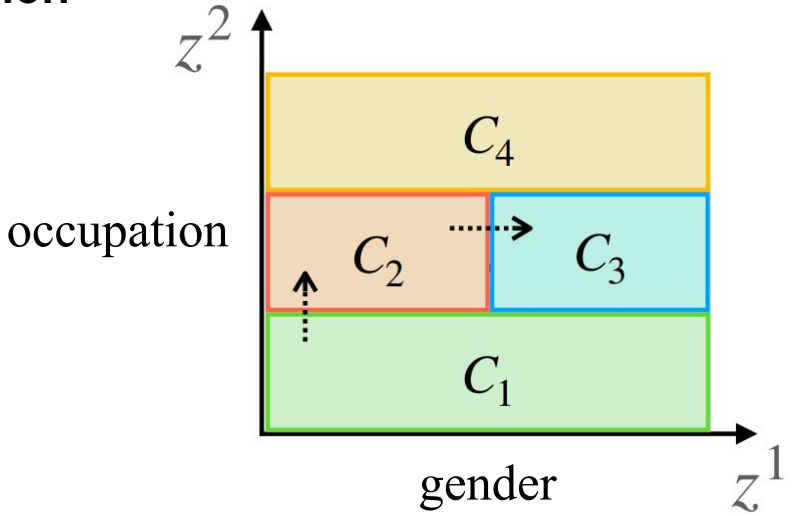
# Example: Income Prediction using LMs

- Each prompt $\mathbf{x}_n$ consists of a candidate's **bio** and an instruction asking whether the candidate earns a **six-figure salary**
- $p(\text{yes}|\mathbf{x}_n)$ is **high** for bios with **high-income** occupations and **low** for bios with to **low-income** occupations
- For **medium-income** occupations, the model's behavior varies by **gender**, leading to **higher** $p(\text{yes}|\mathbf{x}_n)$ for **male** bios and **lower** $p(\text{yes}|\mathbf{x}_n)$ for **female** ones.



# Example: Income Prediction using LMs

- The different ranges of $p(\text{yes}|\mathbf{x}_n)$ give rise to **four** distinct **behavior classes *C***
- The candidate's **gender** and **occupation** are the **"concepts"** that mediate (influence) this behavior



# Challenge: Entanglement of Concepts

- Directly interpreting **individual neurons** or **embedding dimensions** as potential concepts is **not ideal** because LM activations generally **entangle** such concepts
- **Entanglement: Compressing and mixing** multiple concepts into the exact same neuron or embedding dimension
- This necessitates a mapping that effectively **“inverts”** activations back into the space of **disentangled concepts.** This is called **Disentanglement**

# Challenge: Learning Concepts

- Learning concepts using **fully unsupervised learning** is **not identifiable,** meaning there are **infinitely** many correct solutions (entanglements)
- Previous unsupervised methods like **Sparse Autoencoders (SAEs)** project the activations into a massive latent space and use L1 regularization to enforce sparsity on the outputs
- However, SAEs require extensive **post hoc analysis** to interpret individual features, identify which features affect model behavior, and are not able to uniquely recover many concepts (**feature splitting**)

# Problem Setting

- Language model takes in an input sequence $\mathbf{x} \in \mathbf{X}$ and outputs a sequence $\mathbf{y} \in \mathbf{Y}$.
- The fine-grained responses $\boldsymbol{y}$ are categorized into $\boldsymbol{m}$ coarse-grained behavior classes {1, . . . , m} so that each input $\boldsymbol{x}$ is associated with a discrete behavior label $\boldsymbol{c}$.
- For example, **all** queries **x** to the model that elicit responses such as “I am sorry...”, or “I cannot respond...” are mapped to the **same** refusal behavior class.
- Goal is to map model activations $\boldsymbol{g}(\mathbf{x})$ to features $\hat{\mathbf{z}} = \boldsymbol{\tau}(\boldsymbol{g}(\mathbf{x}))$, via a learned encoder $\tau$ , so that the learned features $\hat{\mathbf{z}}$ align with the true mediating concepts **z** (e.g., gender and occupation).

# Learning Causal Differentiating Concepts

- **Assumption 1**

  The true conditional probability of a behavior given the mediating concepts $\mathbf{z}$ encoded by the high-level model $c = \mathcal{M}(\mathbf{x})$ is:

$$p(c|\mathbf{z}) = \frac{\exp(\mathbf{w}_c^T \mathbf{z})}{\sum_{j=1}^{m} \exp(\mathbf{w}_j^T \mathbf{z})}$$

  That is, the model's behavior **c** for an input **x** is related to the mediating latent variables $\mathbf{z}$ by a **logit-linear function**.

# Learning Causal Differentiating Concepts

**Definition 1.** For two inputs $\mathbf{x}_k \in C_k$ and $\mathbf{x}_l \in C_l$ such that $k \neq l$, an **interchange intervention** on a subset of mediating concepts $S \subset \{1, \ldots, d\}$ in the high-level model is defined as $\mathcal{M}_{\mathbf{z}_k^S \leftarrow \mathbf{z}_l^S}(\mathbf{x}_k)$ where for $r \in S$, the $r$-th component of the high-level representation $\mathbf{z}_k$ (associated with the input $\mathbf{x}_k$) is replaced with the corresponding value from $\mathbf{z}_l$ from $\mathbf{x}_l$

**Definition 2.** A set of latent mediating factors $\mathbf{z}^S$ is defined as **causal differentiating concepts** between two groups $C_k$ and $C_l$ if, for some inputs $\mathbf{x}$ that are labeled with class $k$, an interchange intervention on $S$ is both necessary and sufficient to change the behavioral label from $k$ to $l$.

(*Necessary condition*) For any $\mathbf{x}_k \in C_k$ and $\mathbf{x}_l \in C_l$, $\mathcal{M}_{\mathbf{z}_k^{S'} \leftarrow \mathbf{z}_l^{S'}}(\mathbf{x}_k) \neq c_l$, where $\bar{S}' \cap S \neq \varnothing$ (1)

(*Sufficient condition*) For some $\mathbf{x}_k \in C_k$ and $\mathbf{x}_l \in C_l$, $\mathcal{M}_{\mathbf{z}_k^S \leftarrow \mathbf{z}_l^S}(\mathbf{x}_k) = c_l$. (2)

A latent mediating factor $r$ is thus the **1-sparse causal differentiating concept** between a pair of group $C_k$ and $C_l$ if the set of causal mediating concepts between $C_k$ and $C_l$ is $S = \{r\}$.

# Learning Causal Differentiating Concepts

**Assumption 2.** Every dimension $z^r$ is a 1-sparse causal differentiating concept for some pair of groups $C_k$ and $C_l$.

**Corollary 1.** For the groups $C_k$ and $C_l$, consider a 1-sparse latent mask $\boldsymbol{\delta}_{kl} \in \mathbb{R}^d$, which is a $d$-dimensional vector with a nonzero value at position $r$ corresponding to the causal differentiating concept between the two groups and zeros elsewhere. It follows from the necessary and the sufficient conditions of causal differentiating concepts that for some $\mathbf{x}_k \in C_k$ and $\mathbf{x}_l \in C_l$, $\mathbf{z}_l = \mathbf{z}_k + \boldsymbol{\delta}_{kl}$

# Learning Causal Differentiating Concepts

- **Loss Functions:**

$$\min_{\tau,h} \mathbb{E}_{p(\mathbf{x},y)} \left[ -\log h(\tau(g(\mathbf{x})))_y \right] \tag{3}$$

$$\min_{\tau,\delta} \sum_{k,l} \left[ \min_{\substack{\mathbf{x}_i \in C_k \\ \mathbf{x}_j \in C_l}} \mathbb{E}[||\tau(g(\mathbf{x}_j)) - \tau(g(\mathbf{x}_i)) - \hat{\boldsymbol{\delta}}_{kl}||^2] \right] \tag{4}$$

# Learning Causal Differentiating Concepts

- **Theorem 1:** If the *Assumption 1* holds, then the function $\hat{\tau}$ that satisfies *Equation (3)* gives us the true latents (concepts) up to an affine transformation, i.e., $\hat{\mathbf{z}} = A\mathbf{z} + \mathbf{b}$
- **Theorem 2:** If *Assumptions 1* and *2* hold, then the function $\hat{\tau}$ that satisfies *Equations (3) and (4)* identifies the true latents (concepts) up to permutation and scaling. i.e., $A = DP$, where *D* is a diagonal scaling matrix and *P* is a permutation matrix.

# Experiment: Datasets

**Synthetic Data:**

- The **true latent factors *z*** are sampled from a **multivariate Gaussian distribution**, $\mathbf{z} \sim \mathcal{N}(\mu, \sigma) \subset \mathbb{R}^d$ , with *d* = 2 and *d* = 3
- To simulate **input features**, the low-dimensional latent factors ***z*** are projected into a higher-dimensional observation space, $\mathbf{x} \in \mathbb{R}^n$ using various mixing functions:
  - **Linear, non-sparse:** $\mathbf{x} = [2z_0, 5z_1, z_0 + z_1, 2z_0 + z_1, -z_0 + 4z_1, 3z_0 - 2z_1]$
  - **Non-linear, sparse:** $\mathbf{x} = [z_0, 2z_0 + 3z_0^2, 4z_1, 2z_1^2 + z_1^3, 6\sin(z_0), -2\cos(z_1)]$
  - **Non-linear, non-sparse:** $\mathbf{x} = [z_0 z_1, z_0 + 3z_1, z_0^2 z_1, z_0 z_1^3, 2z_0 + z_1, 2z_0 z_1^2]$

# Experiment: Datasets

**Semi-synthetic Data:**

- Took short biographies from the the **BiasBios** dataset, and **manually designed** the function for the probabilites $p(\text{yes} \mid \mathbf{x})$ according the 4 output behavior groups described in the opening Example using **occupation level** and **gender** as the causal factors.

**Non-synthetic Data:**

- Gathered a collection of **harmful** prompts, **harmless** prompts, and **pseudo-harmful** prompts, and extracted $p(\text{refusal} \mid \mathbf{x})$ by summing the probabilities of **refusal tokens** (“I am sorry” and “I cannot”). The probabilities were then clustered into 3 behavior classes.

# Experiment: Evaluation Metrics

- **Disentanglement (D):** Measures the extent to which each **learned** feature ($\hat{\mathbf{z}}_k$) captures *at most* one **true** causal factor ($\mathbf{z}_j$)
- **Completeness (C):** Measures the extent to which each **true** causal factor ($\mathbf{z}_j$) is captured by a *single* **learned** feature ($\hat{\mathbf{z}}_k$)
- **Informativeness (I):** Measures the amount of information that the **learned** vector ($\hat{\mathbf{z}}$) as a whole captures about the **true** factors ($\mathbf{z}$)
- **Mean Correlation Coefficient (MCC):** Pearson correlation between every pair of **learned** and **true** features, followed by 1-to-1 matching

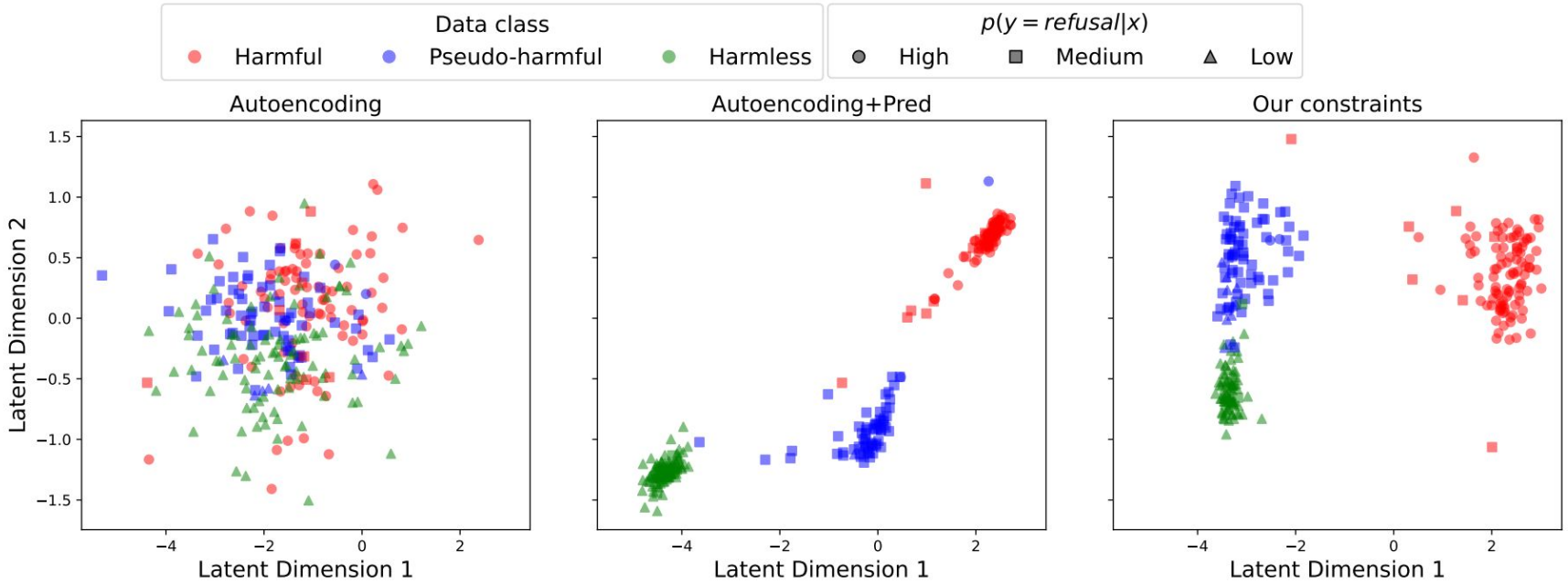
# Results: Synthetic Data

| Mixing fn | Method | MCC | D | C | I |
|---|---|---|---|---|---|
| Linear | Autoencoding | 0.76 | 0.30 | 0.36 | 1.0 |
| | Autoenc+Pred | 0.60 | 0.21 | 0.21 | 1.0 |
| | Our constraints | **0.99** | **1.0** | **1.0** | 1.0 |
| Non-linear, sparse | Autoencoding | 0.81 | 0.24 | 0.33 | 1.0 |
| | Autoenc+Pred | 0.72 | 0.0 | 0.0 | 1.0 |
| | Our constraints | **0.91** | **0.97** | **0.97** | 1.0 |
| Non-linear, non-sparse | Autoencoding | 0.67 | 0.0 | 0.0 | 1.0 |
| | Autoenc+Pred | 0.76 | 0.02 | 0.02 | 1.0 |
| | Our constraints | **0.92** | **0.90** | **0.92** | 1.0 |
| Linear | Autoencoding | 0.77 | 0.34 | 0.35 | 1.0 |
| | Autoenc+Pred | 0.65 | 0.23 | 0.31 | 0.87 |
| | Our constraints | **0.94** | **0.99** | **0.99** | 1.0 |
| Non-linear, sparse | Autoencoding | 0.74 | 0.19 | 0.25 | 1.0 |
| | Autoenc+Pred | 0.77 | 0.54 | 0.51 | 1.0 |
| | Our constraints | **0.90** | **0.92** | **0.95** | 1.0 |
| Non-linear, non-sparse | Autoencoding | 0.78 | 0.51 | 0.60 | 1.0 |
| | Autoenc+Pred | 0.86 | 0.54 | 0.59 | 1.0 |
| | Our constraints | **0.89** | **0.89** | **0.89** | 1.0 |

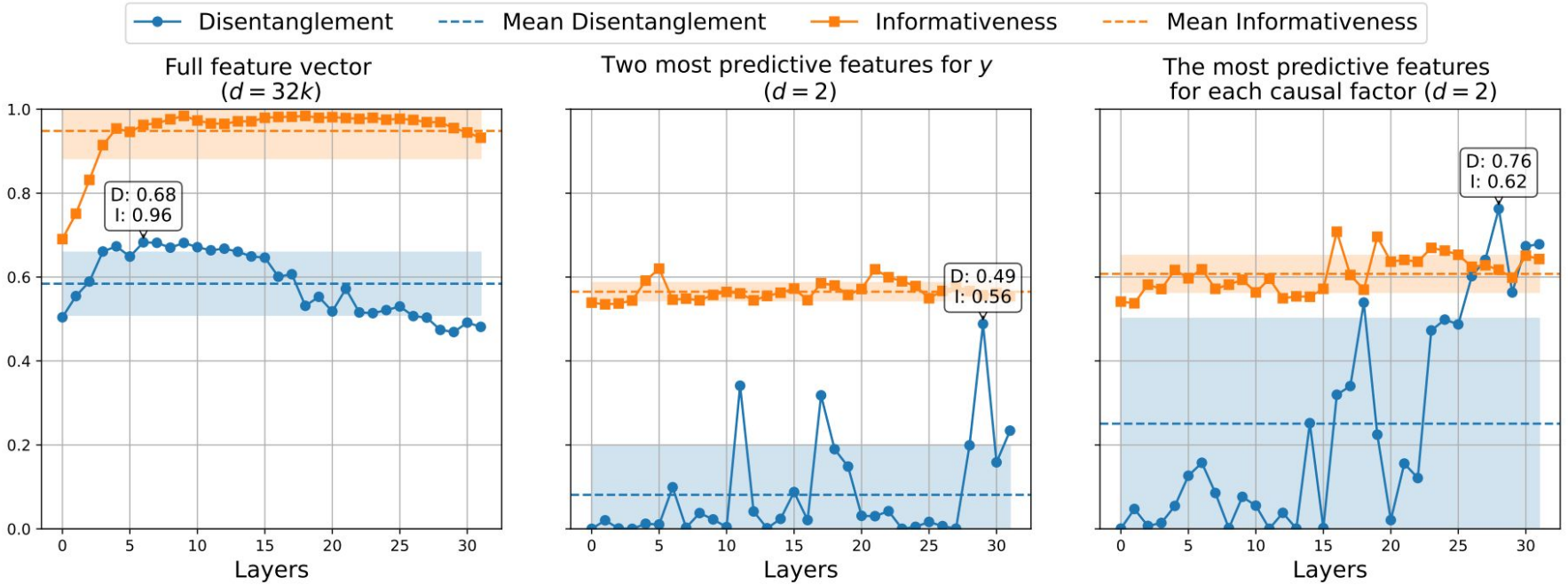
# Results: Semi-synthetic Data

| Model | Method | MCC | D | C | I |
|---|---|---|---|---|---|
| Distilbert-base | Autoencoding | 0.17 | 0.01 | 0.01 | 0.57 |
| | Autoenc+Pred | 0.32 | 0.18 | 0.20 | 0.85 |
| | Our constraints | **0.85** | **0.91** | **0.92** | **0.99** |
| Qwen2-7B | Autoencoding | 0.32 | 0.01 | 0.04 | 0.57 |
| | Autoenc+Pred | 0.69 | 0.06 | 0.06 | 0.83 |
| | Our constraints | **0.79** | **0.80** | **0.84** | **0.97** |
| Llama-3.1-8B | Autoencoding | 0.06 | 0.0 | 0.0 | 0.49 |
| | Autoenc+Pred | 0.54 | 0.05 | 0.11 | 0.84 |
| | Our constraints | **0.81** | **0.86** | **0.88** | **0.98** |
| | Sparse-autoencoders | 0.51 | 0.49 | 0.52 | 0.56 |

# Results: Non-synthetic Data



# Comparison to Sparse Autoencoders



# Sensitivity to Assumptions

**Choice of latent dimensionality:**

- Over-specification (***k*** > true ***d***): Leads to drop in **Completeness (C)**
- Under-specification (***k*** < true ***d***): Leads to drop in **Disentanglement (D)**

**Choice of intervention layer:**

- The proposed constraint applied to **middle layers** can also separate features into disentangled concepts, due to the *linear representation hypothesis*

**Violation of 1-sparsity assumption**

- Further experiments show the results can be also extended to **2-sparse condition,** due to the weaker assumption (that at least *some* pairs have *p*-sparse transition)